



Faculty of Computing

University of Sri Jayewardenepura

CCS3102 | CSE3102 · MOBILE COMPUTING · Group 20



StudyVault

An AI-Powered Study Material Organizer for Mobile

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WHY THIS APP IS NEEDED

The Problem Students Actually Face

1 Materials Are Scattered

PDFs, slides, and photographed handwritten notes spread across WhatsApp, Google Drive and local storage.

2 Disorganized & Duplicated

No centralized structure means poorly named, scattered, duplicated files that are hard to relocate.

3 Critical During Exams

Quick access to the right material matters most exactly when students have the least time to search for it.

4 No Intelligent Understanding

Existing tools (Drive, plain file managers) only store files — none summarize or prioritize content for you in crucial moments.



Project Objectives

Primary Objectives

- Centralized platform to store and organize study material (PDFs & images)
- Enable students to manage files efficiently
- Integrate AI-powered summarization for quick understanding

Secondary Objectives

- Allow students to mark important notes
- Enable secure user authentication
- Reduce time spent searching for and reading materials
- Help students revise using AI-summarized notes and quizzes



BRIDGING PROBLEM TO PRODUCT

Our Solution & Scope

"A centralized vault for academic files, with AI built in to help students understand them faster."



- Platform: Android, built with React Native + Expo
- Satisfies core requirement: functional app, 9 screens, local + cloud data storage
- AI integration is central to the app, not an add-on feature



HOW THE PIECES FIT TOGETHER

System Architecture



React Native / Expo
Frontend

UI, navigation, 9 screens, theming



Firebase
Auth + Firestore

Users, notes & folder metadata



Gemini 2.5 Flash
Google AI API

Summarization + quiz generation



Local Device Storage (expo-file-system)

Actual PDF/image files live here on-device — not inside Firestore. Firestore only stores a relative path reference to each file.



REACT NATIVE + EXPO

Frontend

9 Screens

- ▶ Login
- ▶ Register
- ▶ Home
- ▶ Add Note
- ▶ Note Detail
- ▶ Folder Detail
- ▶ Profile
- ▶ Change Password
- ▶ Quiz



App-Wide Theme System

ThemeContext + AsyncStorage persistence. Every screen uses a createStyles(colors) function — dark/light mode is a real architectural decision, not one hardcoded screen.



Search & Sort

Notes searchable by title, folders by name — ranked so titles starting with the query appear before ones that merely contain it.



Organization Tools

Pin important notes, group into folders, rename notes/folders, duplicate-title prevention.



Backend



Authentication

Firebase Auth — email/password sign-up & login, session persistence, password change with re-authentication.



Cloud Firestore

Notes (title, type, pinned, summarized, summary, createdAt, filePath), folders (name, noteIds[]), and users collections.



Full CRUD

Create, read, update, and delete implemented for notes, folders, and profile — using addDoc, updateDoc, arrayUnion/arrayRemove, deleteDoc.



Local File Storage

expo-file-system manages actual PDF/image bytes on-device, referenced by relative path in Firestore.



AI-Powered Summarization

MODEL

Google Gemini 2.5 Flash
(generateContent REST endpoint)

PURPOSE

Turn an uploaded PDF or image into a structured, exam-useful summary

INPUTS

The uploaded file (base64) + note title

OUTPUTS

Structured JSON: overview, key topics, formulas, worked example

No OCR, No Native Modules

The file is sent as base64 inline_data directly to Gemini, which reads PDFs and images natively in the same call that generates the summary — no separate OCR/extraction step, no native libraries incompatible with Expo Go.

Enforced JSON, Not Markdown

A strict responseSchema forces Gemini to return real JSON (overview, keyTopics[], keyFormulas[], workedExample[]) instead of formatted text. Enforced by the API itself, so the UI renders it directly — no markdown parsing, nothing can render broken.



AI-Generated Revision Quizzes



- Reuses the summary already produced — never re-reads or re-uploads the original file
- Cheaper and faster: one small text-only call instead of a full document call
- Automatically includes numeric/calculation questions when the summary has formulas or worked examples
- Same architectural pattern as summarization (schema-enforced JSON output) — a designed system, not two disconnected features

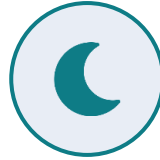


Additional Features



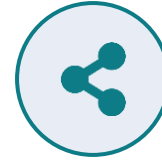
Search files & folders

Search functionality to find files and folders by name



Consistent Theming

Implemented both Light and Dark mode within app to support user friendliness



Share & Download

Export summaries via native share sheet or save-to-device (expo-sharing)



Profile Management

Editable profile, picture upload with cropping, secure account deletion (re-auth required)



Quiz generation

Generate small quiz on the note based on AI summary



PROBLEM-SOLVING, NOT JUST BUILDING

Challenges & How We Solved Them

	Challenge	Solution
Native libraries vs. Expo Go	OCR/PDF-parsing native modules require ejecting from Expo Go.	Send files directly to Gemini as inline data — extraction happens inside the AI call itself.
Hidden “thinking tokens”	Gemini 2.5's internal reasoning tokens were silently eating the visible output budget, truncating results.	Capped <code>thinkingConfig.thinkingBudget</code> and raised <code>maxOutputTokens</code> so summaries/quizzes never got cut off.
Storage cost on the free tier	Storing large binary files in Firestore/Cloud Storage burns through free-tier limits fast.	Hybrid model: metadata in Firestore, actual files kept in local on-device storage.



Live Demonstration

Register/Login → Upload → Organize → Summarize → Quiz → Share



WHERE WE'D TAKE IT NEXT

Future Improvements

1

Push Notifications

Move beyond in-app alerts to real device push notifications

2

Cloud Backup & Sync

Back up local files to the cloud and sync across devices

3

OCR Fallback

Support scanned handwritten notes as a fallback extraction path

4

Cross-Device State Sync

Sync pinned notes and folder structure across a student's devices



Thank you!

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